

Mathematical Methods

Module 2 – Exam Exercise Pack

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at least 100 exam-style exercises with complete guided solutions

How to use this exercise pack

Purpose

This deck trains the techniques of Module 2: sigma-algebras, measures, measurable functions, , convergence theorems, product measures, Fubini-Tonelli and L^p spaces.

- Each exercise is followed by a detailed solution.
- Difficulty is marked from $\star$ to $\star\star\star\star\star$.
- The exercises are designed to be different in structure, not only in numerical values.
- Many solutions include the typical exam strategy: identify the correct theorem, check hypotheses, then compute.

Sigma-algebras and Borel sets

Measures and continuity of measure

Measurable functions

Convergence theorems

Product measures and Fubini-Tonelli

L^p spaces and inequalities

Mixed exam-style review

Module 2

Sigma-algebras and Borel sets

Exercise 1/107 – Check a finite sigma-algebra

Problem

Difficulty: ★

Let $X = \{1, 2, 3, 4\}$ and

$$\mathcal{F} = \{\emptyset, X, \{1, 2\}, \{3, 4\}\}.$$

Decide whether $\mathcal{F}$ is a sigma-algebra on X .

Solution 1/107 – Check a finite sigma-algebra

Detailed solution

We check the three axioms.

- ① $X \in \mathcal{F}$ by definition.
- ② Complements stay in $\mathcal{F}$: $\{1, 2\}^c = \{3, 4\}$, $\{3, 4\}^c = \{1, 2\}$, $X^c = \emptyset$, $\emptyset^c = X$.
- ③ Countable unions stay in $\mathcal{F}$. Since $\mathcal{F}$ has only four sets, any countable union is one of $\emptyset$, $\{1, 2\}$, $\{3, 4\}$ or X .

Therefore $\mathcal{F}$ is a sigma-algebra. It is generated by the partition $\{1, 2\}$, $\{3, 4\}$.

Exercise 2/107 – Find the missing complement

Problem

Difficulty: ★

Let $X = \{a, b, c\}$ and

$$\mathcal{C} = \{\emptyset, X, \{a\}, \{b, c\}, \{b\}\}.$$

Is $\mathcal{C}$ a sigma-algebra?

Solution 2/107 – Find the missing complement

Detailed solution

No. The quickest test is the complement axiom.

$$\{b\} \in \mathcal{C}.$$

If $\mathcal{C}$ were a sigma-algebra, then

$$\{b\}^c = \{a, c\}$$

would also belong to $\mathcal{C}$. But $\{a, c\} \notin \mathcal{C}$. Thus the complement axiom fails, so $\mathcal{C}$ is not a sigma-algebra.

Exercise 3/107 – Sigma-algebra generated by one set

Problem

Difficulty: ★★

Let X be any nonempty set and let $A \subset X$, with $A \neq \emptyset$ and $A \neq X$. Find $\sigma(A)$.

Solution 3/107 – Sigma-algebra generated by one set

Detailed solution

A sigma-algebra containing A must also contain A^c , X and $\emptyset$. Hence

$$\{\emptyset, A, A^c, X\} \subseteq \sigma(A).$$

Conversely, the class

$$\mathcal{G} = \{\emptyset, A, A^c, X\}$$

is a sigma-algebra: complements and countable unions of its elements remain in the same class. Since $\mathcal{G}$ is a sigma-algebra containing A , minimality gives

$$\sigma(A) = \{\emptyset, A, A^c, X\}.$$

Exercise 4/107 – Atoms generated by two sets

Problem

Difficulty: ★★

Let $X = \{1, 2, 3, 4, 5, 6\}$,

$$A = \{1, 2, 3, 4\}, \quad B = \{3, 4, 5\}.$$

Find the atoms of $\sigma(A, B)$ and the number of sets in $\sigma(A, B)$.

Solution 4/107 – Atoms generated by two sets

Detailed solution

The atoms are obtained by all possible intersections of A or A^c with B or B^c :

$$A \cap B = \{3, 4\}, \quad A \cap B^c = \{1, 2\},$$

$$A^c \cap B = \{5\}, \quad A^c \cap B^c = \{6\}.$$

All four atoms are nonempty. Every set in $\sigma(A, B)$ is a union of these atoms. Therefore the number of measurable sets is

$$2^4 = 16.$$

Exercise 5/107 – List a partition-generated sigma-algebra

Problem

Difficulty: ★★

Let $X = \{1, 2, 3, 4, 5\}$ and consider the partition

$$P_1 = \{1\}, \quad P_2 = \{2, 3\}, \quad P_3 = \{4, 5\}.$$

List all sets in the sigma-algebra generated by the partition.

Solution 5/107 – List a partition-generated sigma-algebra

Detailed solution

A partition-generated sigma-algebra contains all unions of the atoms P_1, P_2, P_3 . There are $2^3 = 8$ sets:

$$\emptyset, P_1, P_2, P_3, P_1 \cup P_2, P_1 \cup P_3, P_2 \cup P_3, X.$$

Writing them explicitly:

$$\emptyset, \{1\}, \{2, 3\}, \{4, 5\}, \{1, 2, 3\}, \{1, 4, 5\}, \{2, 3, 4, 5\}, X.$$

Exercise 6/107 – Finite-cofinite algebra

Problem

Difficulty: ★ ★ ★

Let $X = \mathbb{N}$ and let $\mathcal{A}$ be the class of all finite or cofinite subsets of $\mathbb{N}$. Show that $\mathcal{A}$ is an algebra but not a sigma-algebra.

Solution 6/107 – Finite-cofinite algebra

Detailed solution

First, $\mathbb{N}$ is cofinite and $\emptyset$ is finite, so both belong to $\mathcal{A}$. If A is finite, then A^c is cofinite. If A is cofinite, then A^c is finite. Thus $\mathcal{A}$ is closed under complements. The union of two finite sets is finite; the union involving a cofinite set is cofinite. Thus $\mathcal{A}$ is closed under finite unions. So $\mathcal{A}$ is an algebra. It is not a sigma-algebra because

$$\{2\}, \{4\}, \{6\}, \dots \in \mathcal{A},$$

but

$$\bigcup_{n \geq 1} \{2n\} = \{2, 4, 6, \dots\}$$

is neither finite nor cofinite. Hence it is not in $\mathcal{A}$.

Exercise 7/107 – Closure under countable intersections

Problem

Difficulty: ★★

Let $\mathcal{F}$ be a sigma-algebra on X . Prove that if $A_n \in \mathcal{F}$ for all $n \geq 1$, then

$$\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}.$$

Solution 7/107 – Closure under countable intersections

Detailed solution

Use De Morgan's law:

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c.$$

Since $A_n \in \mathcal{F}$, the complement axiom gives $A_n^c \in \mathcal{F}$ for every n . Since $\mathcal{F}$ is closed under countable unions,

$$\bigcup_{n=1}^{\infty} A_n^c \in \mathcal{F}.$$

Taking the complement again gives the desired conclusion.

Exercise 8/107 – Borel status of a complicated set

Problem

Difficulty: ★★

Decide whether the set

$$E = \left(\bigcap_{n=1}^{\infty} (-1/n, 1/n) \right) \cup \mathbb{Q}$$

is Borel in $\mathbb{R}$.

Solution 8/107 – Borel status of a complicated set

Detailed solution

Each interval $(-1/n, 1/n)$ is open, hence Borel. A countable intersection of Borel sets is Borel:

$$\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\}.$$

The set $\mathbb{Q}$ is countable, hence Borel because it is a countable union of singletons. Finally, the union of two Borel sets is Borel. Therefore

$$E \in \mathcal{B}(\mathbb{R}).$$

Exercise 9/107 – Singletons and countable sets

Problem

Difficulty: ★★

Prove that every countable subset $C \subset \mathbb{R}$ is a Borel set.

Solution 9/107 – Singletons and countable sets

Detailed solution

First prove that singletons are Borel. For any $a \in \mathbb{R}$,

$$\{a\} = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, a + \frac{1}{n} \right).$$

Each interval is open, hence Borel. Therefore $\{a\}$ is Borel. If C is countable, write

$$C = \{c_1, c_2, c_3, \dots\}.$$

Then

$$C = \bigcup_{n=1}^{\infty} \{c_n\},$$

a countable union of Borel sets. Hence C is Borel.

Exercise 10/107 – Equivalent generators

Problem

Difficulty: ★ ★ ★

Show that

$$\sigma\left(\{(-\infty, a) : a \in \mathbb{R}\}\right) = \mathcal{B}(\mathbb{R}).$$

Solution 10/107 – Equivalent generators

Detailed solution

Let

$$\mathcal{G} = \sigma\left(\{(-\infty, a) : a \in \mathbb{R}\}\right).$$

First, every $(-\infty, a)$ is open, so it is Borel. Hence $\mathcal{G} \subseteq \mathcal{B}(\mathbb{R})$. For the reverse inclusion, it is enough to show that open intervals belong to $\mathcal{G}$. Since

$$(a, b) = (-\infty, b) \cap (-\infty, a]^c,$$

and

$$(-\infty, a] = \bigcap_{n=1}^{\infty} (-\infty, a + 1/n),$$

we get $(-\infty, a] \in \mathcal{G}$, hence $(a, b) \in \mathcal{G}$. Because $\mathcal{B}(\mathbb{R})$ is generated by open intervals, $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{G}$. Thus the two sigma-algebras coincide.

Exercise 11/107 – A family closed under finite unions only

Problem

Difficulty: ★ ★ ★

Let $X = \mathbb{N}$ and let

$$\mathcal{C} = \{A \subset \mathbb{N} : A \text{ is finite}\} \cup \{\mathbb{N}\}.$$

Is $\mathcal{C}$ a sigma-algebra?

Solution 11/107 – A family closed under finite unions only

Detailed solution

No. Although $\mathcal{C}$ contains $\mathbb{N}$ and is closed under finite unions of finite sets, it fails under complements. For example,

$$\{1\} \in \mathcal{C},$$

but

$$\{1\}^c = \{2, 3, 4, \dots\}$$

is infinite and not equal to $\mathbb{N}$, so it is not in $\mathcal{C}$. Therefore $\mathcal{C}$ is not a sigma-algebra.

Exercise 12/107 – Intersection of sigma-algebras

Problem

Difficulty: ★★ ★

Let $(\mathcal{F}_i)_{i \in I}$ be any family of sigma-algebras on the same set X . Prove that

$$\mathcal{F} = \bigcap_{i \in I} \mathcal{F}_i$$

is a sigma-algebra.

Solution 12/107 – Intersection of sigma-algebras

Detailed solution

We verify the axioms. First, $X \in \mathcal{F}_i$ for every i , hence $X \in \bigcap_i \mathcal{F}_i$. Second, if $A \in \mathcal{F}$, then $A \in \mathcal{F}_i$ for every i . Since each $\mathcal{F}_i$ is a sigma-algebra, $A^c \in \mathcal{F}_i$ for every i . Hence $A^c \in \mathcal{F}$. Third, if $A_n \in \mathcal{F}$ for all n , then for each fixed i we have $A_n \in \mathcal{F}_i$ for all n . Therefore

$$\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}_i$$

for every i . Hence the union belongs to the intersection $\mathcal{F}$. Thus $\mathcal{F}$ is a sigma-algebra.

Module 2

Measures and continuity of measure

Exercise 13/107 – Counting measure on a finite set

Problem

Difficulty: ★

Let $X = \{1, 2, 3, 4, 5\}$ and let μ be counting measure. Compute $\mu(A)$, $\mu(B)$, $\mu(A \cup B)$ and $\mu(A \cap B)$ for

$$A = \{1, 3, 5\}, \quad B = \{2, 3, 4\}.$$

Solution 13/107 – Counting measure on a finite set

Detailed solution

Counting measure gives the number of elements.

$$\mu(A) = 3, \quad \mu(B) = 3.$$

The union is

$$A \cup B = \{1, 2, 3, 4, 5\},$$

so $\mu(A \cup B) = 5$. The intersection is

$$A \cap B = \{3\},$$

so $\mu(A \cap B) = 1$. The inclusion-exclusion check is

$$5 = 3 + 3 - 1.$$

Exercise 14/107 – Weighted counting measure

Problem

Difficulty: ★★

On $X = \{1, 2, 3, 4\}$ define

$$\mu(A) = \sum_{i \in A} w_i, \quad w_1 = 2, w_2 = 1, w_3 = 4, w_4 = 3.$$

Compute $\mu(\{1, 3\})$ and $\mu(X \setminus \{2\})$.

Solution 14/107 – Weighted counting measure

Detailed solution

By definition,

$$\mu(\{1, 3\}) = w_1 + w_3 = 2 + 4 = 6.$$

Also,

$$X \setminus \{2\} = \{1, 3, 4\},$$

so

$$\mu(X \setminus \{2\}) = w_1 + w_3 + w_4 = 2 + 4 + 3 = 9.$$

The measure is additive because it sums nonnegative weights over disjoint elements.

Exercise 15/107 – Dirac measure

Problem

Difficulty: ★★

Let δ_2 be the Dirac measure at 2 on $\mathbb{R}$. Compute

$$\delta_2((1, 3)), \quad \delta_2((2, 5)), \quad \delta_2(\{0, 1\}).$$

Solution 15/107 – Dirac measure

Detailed solution

A Dirac measure only checks whether the distinguished point belongs to the set:

$$\delta_2(A) = \begin{cases} 1, & 2 \in A, \\ 0, & 2 \notin A. \end{cases}$$

Now $2 \in (1, 3)$, so

$$\delta_2((1, 3)) = 1.$$

But $2 \notin (2, 5)$ because the interval is open at 2, so

$$\delta_2((2, 5)) = 0.$$

Finally, $2 \notin \{0, 1\}$, hence

$$\delta_2(\{0, 1\}) = 0.$$

Exercise 16/107 – Lebesgue measure of overlapping intervals

Problem

Difficulty: ★★

Compute the Lebesgue measure of

$$E = (0, 2) \cup (1, 5) \cup (7, 8).$$

Solution 16/107 – Lebesgue measure of overlapping intervals

Detailed solution

First merge overlapping intervals. Since $(0, 2)$ and $(1, 5)$ overlap,

$$(0, 2) \cup (1, 5) = (0, 5).$$

The interval $(7, 8)$ is disjoint from $(0, 5)$. Therefore

$$\lambda(E) = \lambda((0, 5)) + \lambda((7, 8)) = 5 + 1 = 6.$$

The endpoints do not affect Lebesgue measure.

Exercise 17/107 – Continuity from below

Problem

Difficulty: ★★

Let $A_n = [0, 1 - 1/n]$ for $n \geq 2$. Find $\bigcup_{n \geq 2} A_n$ and compute

$$\lim_{n \rightarrow \infty} \lambda(A_n).$$

Solution 17/107 – Continuity from below

Detailed solution

The sets are increasing:

$$A_2 \subset A_3 \subset \cdots.$$

Their union is

$$\bigcup_{n \geq 2} [0, 1 - 1/n] = [0, 1).$$

The measures are

$$\lambda(A_n) = 1 - \frac{1}{n}.$$

Therefore

$$\lim_{n \rightarrow \infty} \lambda(A_n) = 1 = \lambda([0, 1)).$$

This is continuity from below.

Exercise 18/107 – Continuity from above

Problem

Difficulty: ★★

Let $A_n = (0, 1/n)$ in $\mathbb{R}$. Find $\bigcap_{n=1}^{\infty} A_n$ and compute $\lim_n \lambda(A_n)$.

Solution 18/107 – Continuity from above

Detailed solution

The sets are decreasing:

$$(0, 1) \supset (0, 1/2) \supset (0, 1/3) \supset \dots$$

No positive x belongs to all intervals, and 0 is not included. Hence

$$\bigcap_{n=1}^{\infty} A_n = \emptyset.$$

Moreover

$$\lambda(A_n) = \frac{1}{n} \rightarrow 0.$$

Thus

$$\lambda(A_n) \downarrow \lambda(\emptyset) = 0.$$

The finite-measure assumption holds because $\lambda(A_1) = 1 < \infty$.

Exercise 19/107 – Why finite measure is needed

Problem

Difficulty: ★ ★ ★

Give a counterexample showing that continuity from above may fail if $\mu(A_1) = \infty$.

Solution 19/107 – Why finite measure is needed

Detailed solution

Take Lebesgue measure on $\mathbb{R}$ and

$$A_n = (n, \infty).$$

Then

$$A_1 \supset A_2 \supset A_3 \supset \dots$$

and

$$\bigcap_{n=1}^{\infty} A_n = \emptyset.$$

However,

$$\lambda(A_n) = \infty$$

for every n . Hence $\lambda(A_n)$ does not decrease to $0 = \lambda(\emptyset)$. The assumption $\mu(A_1) < \infty$ is essential.

Exercise 20/107 – Subadditivity estimate

Problem

Difficulty: ★★

Let (A_n) be measurable sets such that $\mu(A_n) \leq 2^{-n}$ for every $n \geq 1$. Show that

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) \leq 1.$$

Solution 20/107 – Subadditivity estimate

Detailed solution

Use countable subadditivity:

$$\mu \left(\bigcup_{n=1}^{\infty} A_n \right) \leq \sum_{n=1}^{\infty} \mu(A_n).$$

By the assumption,

$$\sum_{n=1}^{\infty} \mu(A_n) \leq \sum_{n=1}^{\infty} 2^{-n} = 1.$$

Therefore

$$\boxed{\mu \left(\bigcup_{n=1}^{\infty} A_n \right) \leq 1.}$$

Exercise 21/107 – Inclusion-exclusion

Problem

Difficulty: ★★

Suppose $\mu(A) = 7$, $\mu(B) = 5$ and $\mu(A \cap B) = 2$. Assume all measures are finite. Compute $\mu(A \cup B)$ and $\mu(A \setminus B)$.

Solution 21/107 – Inclusion-exclusion

Detailed solution

By inclusion-exclusion,

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) = 7 + 5 - 2 = 10.$$

Also,

$$A = (A \setminus B) \cup (A \cap B)$$

as a disjoint union. Hence

$$\mu(A) = \mu(A \setminus B) + \mu(A \cap B).$$

Therefore

$$\mu(A \setminus B) = 7 - 2 = 5.$$

Exercise 22/107 – Is this a measure?

Problem

Difficulty: ★ ★ ★

On $X = \{1, 2, 3\}$ define

$$\mu(A) = \#A^2.$$

Is μ a measure on $\mathcal{P}(X)$?

Solution 22/107 – Is this a measure?

Detailed solution

No. A measure must be additive over disjoint sets. Take

$$A = \{1\}, \quad B = \{2\}.$$

They are disjoint. Then

$$\mu(A) = 1^2 = 1, \quad \mu(B) = 1^2 = 1.$$

But

$$\mu(A \cup B) = \mu(\{1, 2\}) = 2^2 = 4.$$

Countable additivity would require $4 = 1 + 1$, which is false. Hence μ is not a measure.

Exercise 23/107 – Measure with density

Problem

Difficulty: ★ ★ ★

On $[0, 1]$ define

$$\mu(A) = \int_A 2x \, dx.$$

Compute $\mu([0, 1/2])$ and $\mu((1/2, 1])$.

Solution 23/107 – Measure with density

Detailed solution

The measure is obtained by integrating the density $2x$ over the set. First,

$$\mu([0, 1/2]) = \int_0^{1/2} 2x \, dx = [x^2]_0^{1/2} = \frac{1}{4}.$$

Second,

$$\mu((1/2, 1]) = \int_{1/2}^1 2x \, dx = [x^2]_{1/2}^1 = 1 - \frac{1}{4} = \frac{3}{4}.$$

The endpoint convention does not change the value because singletons have Lebesgue measure zero.

Exercise 24/107 – Countable sets have zero length

Problem

Difficulty: ★★

Show that $\lambda(\mathbb{Q} \cap [0, 1]) = 0$.

Solution 24/107 – Countable sets have zero length

Detailed solution

The set $\mathbb{Q} \cap [0, 1]$ is countable, so enumerate it as

$$\mathbb{Q} \cap [0, 1] = \{q_1, q_2, q_3, \dots\}.$$

Each singleton has Lebesgue measure zero:

$$\lambda(\{q_n\}) = 0.$$

By countable subadditivity,

$$\lambda(\mathbb{Q} \cap [0, 1]) \leq \sum_{n=1}^{\infty} \lambda(\{q_n\}) = 0.$$

Since measures are nonnegative, the measure must be exactly zero.

Exercise 25/107 – Sigma-finiteness

Problem

Difficulty: ★ ★ ★

Show that Lebesgue measure on $\mathbb{R}$ is sigma-finite but not finite.

Solution 25/107 – Sigma-finiteness

Detailed solution

Lebesgue measure is not finite because

$$\lambda(\mathbb{R}) = \infty.$$

To prove sigma-finiteness, write

$$\mathbb{R} = \bigcup_{n=1}^{\infty} [-n, n].$$

Each set $[-n, n]$ is measurable and has finite measure:

$$\lambda([-n, n]) = 2n < \infty.$$

Thus $\mathbb{R}$ is covered by countably many finite-measure sets. Therefore Lebesgue measure is sigma-finite.

Exercise 26/107 – Finite measure from total mass

Problem

Difficulty: ★★

Let μ be a measure on $(X, \mathcal{F})$ with $\mu(X) = 10$. Show that every measurable set A has finite measure and compute $\mu(A^c)$ in terms of $\mu(A)$.

Solution 26/107 – Finite measure from total mass

Detailed solution

Since $A \subseteq X$, monotonicity gives

$$0 \leq \mu(A) \leq \mu(X) = 10.$$

Thus every measurable set has finite measure. Moreover,

$$X = A \cup A^c$$

and the union is disjoint. Hence

$$\mu(X) = \mu(A) + \mu(A^c).$$

Therefore

$$\mu(A^c) = 10 - \mu(A).$$

Module 2

Measurable functions

Exercise 27/107 – Indicator measurability

Problem

Difficulty: ★

Let $(X, \mathcal{F})$ be a measurable space. Prove that $\mathbf{1}_A$ is measurable if and only if $A \in \mathcal{F}$.

Solution 27/107 – Indicator measurability

Detailed solution

If $\mathbf{1}_A$ is measurable, then

$$A = \{x : \mathbf{1}_A(x) > 1/2\} = \mathbf{1}_A^{-1}((1/2, \infty))$$

is measurable. Thus $A \in \mathcal{F}$. Conversely, suppose $A \in \mathcal{F}$. For any $a \in \mathbb{R}$, the set $\{\mathbf{1}_A > a\}$ is one of

$$X, A, \emptyset,$$

depending on the value of a . All these sets belong to $\mathcal{F}$. Therefore $\mathbf{1}_A$ is measurable.

Exercise 28/107 – Simple function measurability

Problem

Difficulty: ★★

Let $A, B \in \mathcal{F}$ and define

$$f = 3\mathbf{1}_A - 2\mathbf{1}_B.$$

Prove that f is measurable.

Solution 28/107 – Simple function measurability

Detailed solution

Since A and B are measurable, the indicators $\mathbf{1}_A$ and $\mathbf{1}_B$ are measurable. A constant multiple of a measurable function is measurable, and the sum of measurable functions is measurable. Therefore

$$f = 3\mathbf{1}_A + (-2)\mathbf{1}_B$$

is measurable. Alternatively, f takes only finitely many values, and each level set can be written using finite combinations of A , B , A^c , B^c .

Exercise 29/107 – Preimage calculation

Problem

Difficulty: ★★

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$. Compute

$$f^{-1}((1, 4])$$

and conclude that this preimage is Borel.

Solution 29/107 – Preimage calculation

Detailed solution

We solve

$$1 < x^2 \leq 4.$$

The inequality $x^2 \leq 4$ gives $-2 \leq x \leq 2$. The inequality $1 < x^2$ gives $x < -1$ or $x > 1$. Combining them,

$$f^{-1}((1, 4]) = [-2, -1) \cup (1, 2].$$

This is a union of intervals, hence it is a Borel set.

Exercise 30/107 – Continuous implies measurable

Problem

Difficulty: ★★

Explain why the function

$$f(x) = \frac{x^2 + 1}{1 + x^4}$$

is Borel measurable on $\mathbb{R}$.

Solution 30/107 – Continuous implies measurable

Detailed solution

The polynomial functions $x^2 + 1$ and $1 + x^4$ are continuous. The denominator satisfies

$$1 + x^4 > 0$$

for all x , so the quotient is continuous on all of $\mathbb{R}$. Every continuous function from $\mathbb{R}$ to $\mathbb{R}$ is Borel measurable. Therefore f is Borel measurable.

Exercise 31/107 – Piecewise rational/irrational function

Problem

Difficulty: ★★

Define

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Is f Borel measurable?

Solution 31/107 – Piecewise rational/irrational function

Detailed solution

Write

$$f(x) = x\mathbf{1}_{\mathbb{Q}}(x).$$

The function $x \mapsto x$ is continuous, hence Borel measurable. The set $\mathbb{Q}$ is countable, hence Borel. Therefore $\mathbf{1}_{\mathbb{Q}}$ is measurable. The product of measurable functions is measurable. Hence f is Borel measurable. Notice that measurability is much weaker than continuity.

Exercise 32/107 – Operations preserve measurability

Problem

Difficulty: ★★

Let $f, g : X \rightarrow \mathbb{R}$ be measurable. Prove that

$$h(x) = \max\{f(x), g(x)\}$$

is measurable.

Solution 32/107 – Operations preserve measurability

Detailed solution

Use the preimage characterization. For any $a \in \mathbb{R}$,

$$\{x : h(x) > a\} = \{x : \max(f(x), g(x)) > a\}.$$

This happens exactly when $f(x) > a$ or $g(x) > a$:

$$\{h > a\} = \{f > a\} \cup \{g > a\}.$$

Since f and g are measurable, both sets on the right belong to $\mathcal{F}$. Their union also belongs to $\mathcal{F}$. Thus h is measurable.

Exercise 33/107 – Limit of measurable functions

Problem

Difficulty: ★★

Let (f_n) be measurable and suppose $f_n(x) \rightarrow f(x)$ for every x . Prove that f is measurable.

Solution 33/107 – Limit of measurable functions

Detailed solution

Use the identity

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \limsup_{n \rightarrow \infty} f_n(x).$$

For every $a \in \mathbb{R}$,

$$\{f > a\} = \bigcup_{m=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n \geq N} \{f_n > a + 1/m\}.$$

Each set $\{f_n > a + 1/m\}$ is measurable. Countable unions and intersections preserve measurability. Therefore $\{f > a\}$ is measurable for every a , so f is measurable.

Exercise 34/107 – Supremum of a sequence

Problem

Difficulty: ★ ★ ★

Let (f_n) be measurable and define

$$g(x) = \sup_{n \geq 1} f_n(x).$$

Prove that g is measurable.

Solution 34/107 – Supremum of a sequence

Detailed solution

For any $a \in \mathbb{R}$,

$$\{g > a\} = \left\{ \sup_{n \geq 1} f_n > a \right\}.$$

The supremum is greater than a if and only if at least one f_n is greater than a :

$$\{g > a\} = \bigcup_{n=1}^{\infty} \{f_n > a\}.$$

Each set $\{f_n > a\}$ is measurable. A countable union of measurable sets is measurable. Hence g is measurable.

Exercise 35/107 – A nonmeasurable indicator

Problem

Difficulty: ★ ★ ★

Let $(X, \mathcal{F})$ be a measurable space and let $A \subset X$ with $A \notin \mathcal{F}$. Show that $\mathbf{1}_A$ is not measurable.

Solution 35/107 – A nonmeasurable indicator

Detailed solution

Assume, by contradiction, that $\mathbf{1}_A$ is measurable. Then the preimage of the open interval $(1/2, \infty)$ must be measurable:

$$\mathbf{1}_A^{-1}((1/2, \infty)) \in \mathcal{F}.$$

But

$$\mathbf{1}_A^{-1}((1/2, \infty)) = A.$$

This would imply $A \in \mathcal{F}$, contradicting the assumption. Therefore $\mathbf{1}_A$ is not measurable.

Exercise 36/107 – Image versus preimage

Problem

Difficulty: ★ ★ ★

A student says: “If f is measurable, then $f(A)$ is Borel for every measurable A .” Explain why this is not the definition and what measurability actually requires.

Solution 36/107 – Image versus preimage

Detailed solution

Measurability is defined through preimages, not images. A function $f : (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is measurable if

$$f^{-1}(B) \in \mathcal{F}$$

for every Borel set $B \subset \mathbb{R}$. The condition concerns sets in the target pulled back to the domain. It does not require that images of measurable sets are Borel. The correct exam strategy is therefore always: start from a Borel set in the codomain and compute its preimage.

Exercise 37/107 – Measurable step function from level sets

Problem

Difficulty: ★ ★ ★

Let

$$f = 2\mathbf{1}_A + 5\mathbf{1}_B + 7\mathbf{1}_C,$$

where $A, B, C \in \mathcal{F}$ are pairwise disjoint. Describe $\{f > 4\}$ and $\{f = 0\}$.

Solution 37/107 – Measurable step function from level sets

Detailed solution

Because the sets are pairwise disjoint, the function takes value 2 on A , value 5 on B , value 7 on C , and value 0 outside $A \cup B \cup C$. Thus

$$\{f > 4\} = B \cup C.$$

Also,

$$\{f = 0\} = X \setminus (A \cup B \cup C).$$

Both sets are measurable because $\mathcal{F}$ is closed under finite unions and complements.

Exercise 38/107 – Measurability of a monotone function

Problem

Difficulty: ★ ★ ★★

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be increasing. Prove that f is Borel measurable.

Solution 38/107 – Measurability of a monotone function

Detailed solution

It is enough to show that $\{x : f(x) > a\}$ is Borel for every $a \in \mathbb{R}$. Since f is increasing, the set

$$E_a = \{x : f(x) > a\}$$

is an upper interval: if $x \in E_a$ and $y > x$, then $f(y) \geq f(x) > a$, so $y \in E_a$. Therefore E_a has one of the forms

$$\emptyset, \quad \mathbb{R}, \quad (c, \infty), \quad [c, \infty)$$

for some $c \in \mathbb{R}$. All these sets are Borel. Hence f is Borel measurable.

Exercise 39/107 – Measurable absolute value

Problem

Difficulty: ★★

Let f be measurable. Prove that $|f|$ is measurable.

Solution 39/107 – Measurable absolute value

Detailed solution

For any $a \in \mathbb{R}$, if $a < 0$, then

$$\{|f| > a\} = X,$$

which is measurable. If $a \geq 0$, then

$$\{|f| > a\} = \{f > a\} \cup \{f < -a\}.$$

Both sets are measurable because f is measurable. Their union is measurable. Therefore $|f|$ is measurable.

Module 2

Exercise 40/107 – Integral of a simple function

Problem

Difficulty: ★

Let A, B be disjoint measurable sets with $\mu(A) = 3$ and $\mu(B) = 4$. Compute

$$\int (2\mathbf{1}_A + 5\mathbf{1}_B) d\mu.$$

Solution 40/107 – Integral of a simple function

Detailed solution

For a nonnegative simple function with disjoint level sets,

$$\int (2\mathbf{1}_A + 5\mathbf{1}_B) d\mu = 2\mu(A) + 5\mu(B).$$

Substitute the values:

$$2 \cdot 3 + 5 \cdot 4 = 6 + 20 = 26.$$

Thus the integral equals

$$\boxed{26}.$$

Exercise 41/107 – Simple function with overlapping sets

Problem

Difficulty: ★ ★ ★

Let $\mu(A) = 4$, $\mu(B) = 5$ and $\mu(A \cap B) = 2$. Compute

$$\int (\mathbf{1}_A + 3\mathbf{1}_B) d\mu.$$

Solution 41/107 – Simple function with overlapping sets

Detailed solution

Linearity of the integral gives the fastest solution:

$$\int (\mathbf{1}_A + 3\mathbf{1}_B) d\mu = \int \mathbf{1}_A d\mu + 3 \int \mathbf{1}_B d\mu.$$

Since $\int \mathbf{1}_A d\mu = \mu(A)$ and $\int \mathbf{1}_B d\mu = \mu(B)$,

$$\int (\mathbf{1}_A + 3\mathbf{1}_B) d\mu = 4 + 3 \cdot 5 = 19.$$

The overlap does not cause double-counting problems because the function really equals 4 on $A \cap B$, and linearity already accounts for this.

Exercise 42/107 – Integral under counting measure

Problem

Difficulty: ★★

On $\mathbb{N}$ with counting measure, compute

$$\int_{\mathbb{N}} \frac{1}{n^2} d\mu(n).$$

Solution 42/107 – Integral under counting measure

Detailed solution

Integration with respect to counting measure is summation:

$$\int_{\mathbb{N}} \frac{1}{n^2} d\mu(n) = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

The series is the classical p -series with $p = 2$, hence it converges. Its exact value is

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Therefore the integral is finite and equals $\pi^2/6$.

Exercise 43/107 – Integral with a Dirac measure

Problem

Difficulty: ★★

Let δ_3 be the Dirac measure at 3 on $\mathbb{R}$. Compute

$$\int_{\mathbb{R}} (x^2 - 4x + 1) d\delta_3(x).$$

Solution 43/107 – Integral with a Dirac measure

Detailed solution

Integration with respect to a Dirac measure evaluates the function at the distinguished point:

$$\int f d\delta_3 = f(3).$$

Here

$$f(x) = x^2 - 4x + 1.$$

Thus

$$f(3) = 9 - 12 + 1 = -2.$$

Therefore

$$\int_{\mathbb{R}} (x^2 - 4x + 1) d\delta_3(x) = -2.$$

Exercise 44/107 – Integral of an indicator

Problem

Difficulty: ★

Compute

$$\int_{\mathbb{R}} \mathbf{1}_{[2,5)}(x) d\lambda(x).$$

Solution 44/107 – Integral of an indicator

Detailed solution

The integral of an indicator is the measure of the set:

$$\int_{\mathbb{R}} \mathbf{1}_{[2,5)} d\lambda = \lambda([2, 5)).$$

The length is

$$5 - 2 = 3.$$

Endpoints do not affect Lebesgue measure. Therefore the integral equals 3.

Exercise 45/107 – Integrability near zero

Problem

Difficulty: ★ ★ ★

For which values of $a \in \mathbb{R}$ is

$$f(x) = x^{-a} \mathbf{1}_{(0,1)}(x)$$

Lebesgue integrable on $(0, 1)$?

Solution 45/107 – Integrability near zero

Detailed solution

We compute the improper integral:

$$\int_0^1 x^{-a} dx.$$

If $a \neq 1$,

$$\int_0^1 x^{-a} dx = \left. \frac{x^{1-a}}{1-a} \right|_0^1.$$

This is finite exactly when $1 - a > 0$, i.e.

$$a < 1.$$

If $a = 1$, the integral is

$$\int_0^1 \frac{1}{x} dx = \infty.$$

Thus

$$f \in L^1(0,1) \iff a < 1.$$

Exercise 46/107 – Integrability at infinity

Problem

Difficulty: ★ ★ ★

For which $p > 0$ is

$$f(x) = \frac{1}{1+x^p}$$

in $L^1((0, \infty))$?

Solution 46/107 – Integrability at infinity

Detailed solution

Near 0, the function is bounded by 1, so there is no problem. At infinity,

$$\frac{1}{1+x^p} \sim \frac{1}{x^p}.$$

The integral

$$\int_1^{\infty} x^{-p} dx$$

converges if and only if $p > 1$. Therefore

$$\boxed{\frac{1}{1+x^p} \in L^1((0, \infty)) \iff p > 1.}$$

Exercise 47/107 – Positive and negative parts

Problem

Difficulty: ★ ★ ★

Let

$$f(x) = 2\mathbf{1}_A(x) - 3\mathbf{1}_B(x),$$

where $A \cap B = \emptyset$, $\mu(A) = 4$ and $\mu(B) = 1$. Find f^+ , f^- and $\int f \, d\mu$.

Solution 47/107 – Positive and negative parts

Detailed solution

The positive part is

$$f^+ = \max(f, 0) = 2\mathbf{1}_A.$$

The negative part is

$$f^- = \max(-f, 0) = 3\mathbf{1}_B.$$

Then

$$\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu.$$

Compute:

$$\int f^+ \, d\mu = 2\mu(A) = 8, \quad \int f^- \, d\mu = 3\mu(B) = 3.$$

Thus

$$\int f \, d\mu = 8 - 3 = 5.$$

Exercise 48/107 – Integral of the Dirichlet function

Problem

Difficulty: ★★

Let

$$f(x) = \mathbf{1}_{\mathbb{Q} \cap [0,1]}(x).$$

Compute the Lebesgue integral of f over $[0, 1]$.

Solution 48/107 – Integral of the Dirichlet function

Detailed solution

The function is the indicator of $\mathbb{Q} \cap [0, 1]$. Therefore

$$\int_{[0,1]} f \, d\lambda = \lambda(\mathbb{Q} \cap [0, 1]).$$

The set $\mathbb{Q} \cap [0, 1]$ is countable, so it has Lebesgue measure zero. Hence

$$\int_{[0,1]} \mathbf{1}_{\mathbb{Q} \cap [0,1]} \, d\lambda = 0.$$

This is a typical example where the Lebesgue integral exists although the Riemann integral does not.

Exercise 49/107 – Changing a function on a null set

Problem

Difficulty: ★★

Let $f(x) = x$ on $[0, 1]$ and define g by

$$g(x) = \begin{cases} 100, & x = 1/2, \\ x, & x \neq 1/2. \end{cases}$$

Compute $\int_0^1 g(x) dx$.

Solution 49/107 – Changing a function on a null set

Detailed solution

The functions f and g differ only at the single point $\{1/2\}$. A singleton has Lebesgue measure zero. Changing a function on a null set does not change its Lebesgue integral. Thus

$$\int_0^1 g(x) dx = \int_0^1 x dx.$$

Compute:

$$\int_0^1 x dx = \frac{1}{2}.$$

Therefore the integral is $1/2$.

Exercise 50/107 – Absolute integrability

Problem

Difficulty: ★ ★ ★

Let

$$f(x) = \frac{\sin x}{x} \mathbf{1}_{(1, \infty)}(x).$$

Is it enough to know that $\int_1^\infty f(x) dx$ converges to conclude that $f \in L^1$?

Solution 50/107 – Absolute integrability

Detailed solution

No. Membership in L^1 requires

$$\int_1^{\infty} |f(x)| dx < \infty.$$

The integral of $\sin x/x$ converges conditionally by oscillation, but

$$\int_1^{\infty} \frac{|\sin x|}{x} dx = \infty.$$

Thus f is not absolutely integrable. The key exam distinction is:

convergent improper integral $\nrightarrow L^1$ integrability.

Exercise 51/107 – Layer-cake for a simple function

Problem

Difficulty: ★★ ★

Let $f = 3\mathbf{1}_A + \mathbf{1}_B$, where $A \cap B = \emptyset$, $\mu(A) = 2$ and $\mu(B) = 5$. Verify

$$\int f \, d\mu = \int_0^\infty \mu(f > t) \, dt.$$

Solution 51/107 – Layer-cake for a simple function

Detailed solution

First compute directly:

$$\int f d\mu = 3\mu(A) + 1\mu(B) = 3 \cdot 2 + 5 = 11.$$

Now compute distribution sizes. For $0 \leq t < 1$, $f > t$ on $A \cup B$, so

$$\mu(f > t) = 2 + 5 = 7.$$

For $1 \leq t < 3$, $f > t$ only on A , so $\mu(f > t) = 2$. For $t \geq 3$, $\mu(f > t) = 0$. Thus

$$\int_0^\infty \mu(f > t) dt = \int_0^1 7 dt + \int_1^3 2 dt = 7 + 4 = 11.$$

The formula is verified.

Exercise 52/107 – Monotonicity of the integral

Problem

Difficulty: ★★

Let $0 \leq f \leq g$ almost everywhere. Prove that

$$\int f \, d\mu \leq \int g \, d\mu.$$

Solution 52/107 – Monotonicity of the integral

Detailed solution

Since $0 \leq f \leq g$ almost everywhere, the function $g - f$ is nonnegative almost everywhere. The integral of a nonnegative measurable function is nonnegative:

$$\int (g - f) d\mu \geq 0.$$

Using linearity whenever the terms are well-defined,

$$\int g d\mu - \int f d\mu = \int (g - f) d\mu \geq 0.$$

Therefore

$$\int f d\mu \leq \int g d\mu.$$

Exercise 53/107 – Integral of a step approximation

Problem

Difficulty: ★ ★ ★

On $[0, 1]$, define

$$s_n(x) = \sum_{k=1}^n \frac{k-1}{n} \mathbf{1}_{[(k-1)/n, k/n)}(x).$$

Compute $\int_0^1 s_n(x) dx$.

Solution 53/107 – Integral of a step approximation

Detailed solution

The function is constant on each subinterval of length $1/n$. Therefore

$$\int_0^1 s_n(x) dx = \sum_{k=1}^n \frac{k-1}{n} \cdot \frac{1}{n}.$$

Thus

$$\int_0^1 s_n(x) dx = \frac{1}{n^2} \sum_{k=1}^n (k-1) = \frac{1}{n^2} \frac{n(n-1)}{2}.$$

Hence

$$\boxed{\int_0^1 s_n(x) dx = \frac{n-1}{2n}.$$

This tends to $1/2$, the integral of x on $[0, 1]$.

Exercise 54/107 – A function finite a.e. but not integrable

Problem

Difficulty: ★ ★ ★

Give an example of a measurable function on $(0,1)$ that is finite almost everywhere but not integrable.

Solution 54/107 – A function finite a.e. but not integrable

Detailed solution

Take

$$f(x) = \frac{1}{x}, \quad x \in (0, 1).$$

It is finite for every $x \in (0, 1)$ and is measurable because it is continuous on $(0, 1)$. However,

$$\int_0^1 \frac{1}{x} dx = \infty.$$

Therefore f is finite almost everywhere but not in $L^1(0, 1)$. The lesson is: finite a.e. is much weaker than integrable.

Module 2

Convergence theorems

Exercise 55/107 – Monotone convergence with indicators

Problem

Difficulty: ★★

Let $A_n = [0, 1 - 1/n]$ for $n \geq 2$ and $f_n = \mathbf{1}_{A_n}$ on $[0, 1]$. Use the Monotone Convergence Theorem to compute

$$\lim_{n \rightarrow \infty} \int_0^1 f_n \, dx.$$

Solution 55/107 – Monotone convergence with indicators

Detailed solution

The sets A_n are increasing, so the functions $f_n = \mathbf{1}_{A_n}$ increase pointwise. The pointwise limit is

$$f = \mathbf{1}_{[0,1)}.$$

By the Monotone Convergence Theorem,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n \, dx = \int_0^1 f \, dx.$$

Now

$$\int_0^1 \mathbf{1}_{[0,1)} \, dx = \lambda([0,1)) = 1.$$

Therefore the limit equals 1.

Exercise 56/107 – Moving spike counterexample

Problem

Difficulty: ★ ★ ★

On $[0, 1]$, define

$$f_n(x) = n \mathbf{1}_{(0, 1/n)}(x).$$

Find the pointwise limit and compare $\lim_n \int f_n$ with the integral of the limit.

Solution 56/107 – Moving spike counterexample

Detailed solution

For every fixed $x > 0$, eventually $x > 1/n$, so $f_n(x) = 0$. At $x = 0$, the interval is open, so $f_n(0) = 0$. Hence

$$f_n(x) \rightarrow 0 \quad \text{for every } x \in [0, 1].$$

But

$$\int_0^1 f_n(x) dx = n\lambda((0, 1/n)) = 1.$$

Thus

$$\lim_n \int f_n = 1, \quad \int \lim_n f_n = 0.$$

This shows that pointwise convergence alone is not enough to interchange limit and integral.

Exercise 57/107 – Dominated convergence: x^n

Problem

Difficulty: ★★

Let $f_n(x) = x^n$ on $[0, 1]$. Use the Dominated Convergence Theorem to find

$$\lim_{n \rightarrow \infty} \int_0^1 x^n dx.$$

Solution 57/107 – Dominated convergence: x^n

Detailed solution

For every $x \in [0, 1)$, $x^n \rightarrow 0$, while $1^n = 1$. Thus the pointwise limit is 0 except at $x = 1$. The point $\{1\}$ has measure zero, so the limit is 0 almost everywhere. Also,

$$0 \leq x^n \leq 1$$

for all n , and 1 is integrable on $[0, 1]$. By the Dominated Convergence Theorem,

$$\lim_{n \rightarrow \infty} \int_0^1 x^n dx = \int_0^1 0 dx = 0.$$

Indeed, directly $\int_0^1 x^n dx = 1/(n+1)$.

Exercise 58/107 – Dominated convergence with trigonometry

Problem

Difficulty: ★ ★ ★

Show that

$$\lim_{n \rightarrow \infty} \int_0^\pi \frac{\sin(nx)}{n} dx = 0.$$

Use a convergence theorem.

Solution 58/107 – Dominated convergence with trigonometry

Detailed solution

Define

$$f_n(x) = \frac{\sin(nx)}{n}.$$

For every x , $|\sin(nx)| \leq 1$, so

$$|f_n(x)| \leq \frac{1}{n} \leq 1.$$

Also, $f_n(x) \rightarrow 0$ for every x because the numerator is bounded and $n \rightarrow \infty$. The dominating function $g(x) = 1$ is integrable on $[0, \pi]$. By Dominated Convergence,

$$\lim_{n \rightarrow \infty} \int_0^\pi \frac{\sin(nx)}{n} dx = \int_0^\pi 0 dx = 0.$$

Exercise 59/107 – Fatou strict inequality

Problem

Difficulty: ★ ★ ★

Let

$$f_n(x) = n \mathbf{1}_{(0, 1/n)}(x)$$

on $[0, 1]$. Verify that Fatou's inequality can be strict.

Solution 59/107 – Fatou strict inequality

Detailed solution

We already know $f_n \rightarrow 0$ pointwise on $[0, 1]$. Therefore

$$\int_0^1 \liminf_{n \rightarrow \infty} f_n \, dx = \int_0^1 0 \, dx = 0.$$

On the other hand,

$$\int_0^1 f_n \, dx = 1$$

for every n . Hence

$$\liminf_{n \rightarrow \infty} \int_0^1 f_n \, dx = 1.$$

Fatou gives

$$0 \leq 1,$$

and the inequality is strict.

Exercise 60/107 – Bounded convergence on finite measure spaces

Problem

Difficulty: ★ ★ ★

Let $\mu(X) < \infty$. Suppose $f_n \rightarrow f$ pointwise and $|f_n| \leq M$ for all n . Prove that

$$\int f_n d\mu \rightarrow \int f d\mu.$$

Solution 60/107 – Bounded convergence on finite measure spaces

Detailed solution

Since $|f_n| \leq M$, the constant function $g = M$ dominates all f_n :

$$|f_n| \leq g.$$

Because $\mu(X) < \infty$,

$$\int_X g \, d\mu = M\mu(X) < \infty.$$

Thus g is integrable. By the Dominated Convergence Theorem,

$$\lim_{n \rightarrow \infty} \int f_n \, d\mu = \int f \, d\mu.$$

This is the bounded convergence theorem as a special case of DCT.

Exercise 01/107 – Bounded convergence may fail on infinite measure

Problem

Difficulty: ★★

On $\mathbb{R}$ with Lebesgue measure, let

$$f_n = \mathbf{1}_{[n, n+1]}.$$

Show that $f_n \rightarrow 0$ pointwise and $|f_n| \leq 1$, but the integrals do not converge to 0.

Solution 01/107 - Bounded convergence may fail on infinite measure

Detailed solution

Fix $x \in \mathbb{R}$. For sufficiently large n , the interval $[n, n+1]$ is far to the right of x , so $f_n(x) = 0$. Hence $f_n(x) \rightarrow 0$ pointwise. Also $|f_n| \leq 1$. However, the dominating function 1 is not integrable on $\mathbb{R}$:

$$\int_{\mathbb{R}} 1 \, dx = \infty.$$

Moreover,

$$\int_{\mathbb{R}} f_n \, dx = \lambda([n, n+1]) = 1$$

for every n . Thus the integrals do not converge to 0.

Exercise 62/107 – Monotone convergence for a series

Problem

Difficulty: ★★

Let $f_n \geq 0$ be measurable. Show that

$$\int \sum_{n=1}^{\infty} f_n d\mu = \sum_{n=1}^{\infty} \int f_n d\mu.$$

Solution 62/107 – Monotone convergence for a series

Detailed solution

Define partial sums

$$s_N = \sum_{n=1}^N f_n.$$

Because all f_n are nonnegative,

$$s_N \uparrow \sum_{n=1}^{\infty} f_n.$$

By the Monotone Convergence Theorem,

$$\int \sum_{n=1}^{\infty} f_n d\mu = \lim_{N \rightarrow \infty} \int s_N d\mu.$$

By finite linearity,

$$\int s_N d\mu = \sum_{n=1}^N \int f_n d\mu.$$

Taking limits gives the result.

Exercise 63/107 – DCT for a sequence on $\mathbb{N}$

Problem

Difficulty: ★ ★ ★

On $\mathbb{N}$ with counting measure, define

$$f_n(k) = \frac{1}{k^2} \mathbf{1}_{\{k \geq n\}}.$$

Use DCT to compute $\lim_n \sum_{k=1}^{\infty} f_n(k)$.

Solution 63/107 – DCT for a sequence on $\mathbb{N}$

Detailed solution

For each fixed k , eventually $n > k$, so $\mathbf{1}_{\{k \geq n\}} = 0$. Hence $f_n(k) \rightarrow 0$. Also,

$$0 \leq f_n(k) \leq \frac{1}{k^2}.$$

The dominating sequence is summable:

$$\sum_{k=1}^{\infty} \frac{1}{k^2} < \infty.$$

By Dominated Convergence for counting measure,

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{\infty} f_n(k) = \sum_{k=1}^{\infty} 0 = 0.$$

Exercise 64/107 – Check DCT hypotheses

Problem

Difficulty: ★★

Let

$$f_n(x) = \frac{x}{1 + nx^2}$$

on $[0, 1]$. Show that $\int_0^1 f_n(x) dx \rightarrow 0$.

Solution 64/107 – Check DCT hypotheses

Detailed solution

For each fixed x , if $x = 0$ then $f_n(0) = 0$. If $x > 0$, then

$$\frac{x}{1 + nx^2} \rightarrow 0.$$

Thus $f_n \rightarrow 0$ pointwise. Moreover, for $x \in [0, 1]$,

$$0 \leq f_n(x) \leq x \leq 1.$$

The function $g(x) = 1$ is integrable on $[0, 1]$. By DCT,

$$\int_0^1 f_n(x) dx \rightarrow 0.$$

Exercise 65/107 – A missing domination

Problem

Difficulty: ★ ★ ★

Let

$$f_n(x) = n \mathbf{1}_{(0, 1/n)}(x).$$

Explain why no integrable function g can dominate all f_n on $[0, 1]$.

Solution 65/107 – A missing domination

Detailed solution

Suppose $g \in L^1([0, 1])$ and $g \geq f_n$ for all n . For every $x \in (0, 1)$, choose n such that $x < 1/n$, i.e. $n < 1/x$ is not ideal. Instead take infinitely many n only when x is very small. A cleaner argument uses integrals. If $g \geq f_n$, then

$$\int_0^{1/n} g(x) \, dx \geq \int_0^{1/n} n \, dx = 1$$

for every n . But since $g \in L^1$, absolute continuity of the Lebesgue integral implies

$$\int_0^{1/n} g(x) \, dx \rightarrow 0.$$

Contradiction. Hence no integrable dominator exists.

Exercise 66/107 – Fatou lower bound

Problem

Difficulty: ★ ★ ★

Suppose $f_n \geq 0$, $f_n \rightarrow f$ almost everywhere, and

$$\int f_n d\mu \leq 5$$

for all n . What can you say about $\int f d\mu$?

Solution 66/107 – Fatou lower bound

Detailed solution

By Fatou's lemma,

$$\int f \, d\mu = \int \liminf_{n \rightarrow \infty} f_n \, d\mu \leq \liminf_{n \rightarrow \infty} \int f_n \, d\mu.$$

Since each integral is at most 5,

$$\liminf_{n \rightarrow \infty} \int f_n \, d\mu \leq 5.$$

Therefore

$$\boxed{\int f \, d\mu \leq 5.}$$

This is a common compactness-type use of Fatou.

Exercise 67/107 – Increasing functions but infinite limit

Problem

Difficulty: ★★

Let $f_n(x) = n\mathbf{1}_{(0,1)}(x)$ on $(0, 1)$. Apply the Monotone Convergence Theorem.

Solution 67/107 – Increasing functions but infinite limit

Detailed solution

The sequence is increasing:

$$f_n(x) = n \leq n+1 = f_{n+1}(x).$$

The pointwise limit is

$$f(x) = +\infty$$

for every $x \in (0, 1)$. By MCT,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 f(x) dx.$$

The left side is

$$\lim_{n \rightarrow \infty} n = \infty,$$

so the integral of the limit is infinite.

Exercise 68/107 – Almost everywhere convergence is enough

Problem

Difficulty: ★ ★ ★

Let $f_n \rightarrow f$ almost everywhere and suppose $|f_n| \leq g$ with $g \in L^1$. Does DCT still apply?

Solution 68/107 – Almost everywhere convergence is enough

Detailed solution

Yes. The Dominated Convergence Theorem only requires convergence almost everywhere, not necessarily everywhere. There exists a null set N such that $f_n(x) \rightarrow f(x)$ for all $x \notin N$. Changing functions on N does not change their integrals. Thus DCT gives

$$\int |f_n - f| d\mu \rightarrow 0$$

and in particular

$$\int f_n d\mu \rightarrow \int f d\mu.$$

The null set is invisible to the integral.

Exercise 69/107 – Dominated convergence with a parameter

Problem

Difficulty: ★ ★ ★

For $a_n \rightarrow a$, show that

$$\int_0^1 e^{-a_n x} dx \rightarrow \int_0^1 e^{-ax} dx$$

assuming all $a_n \geq 0$.

Solution 69/107 – Dominated convergence with a parameter

Detailed solution

For each $x \in [0, 1]$,

$$e^{-a_n x} \rightarrow e^{-ax}$$

by continuity of the exponential. Since $a_n \geq 0$ and $x \geq 0$,

$$0 \leq e^{-a_n x} \leq 1.$$

The function $g(x) = 1$ is integrable on $[0, 1]$. Therefore, by the Dominated Convergence Theorem,

$$\int_0^1 e^{-a_n x} dx \rightarrow \int_0^1 e^{-ax} dx.$$

Module 2

Product measures and Fubini-Tonelli

Exercise 70/107 – Product measure of a rectangle

Problem

Difficulty: ★

Let $A = [0, 2]$ and $B = (1, 4)$ in $\mathbb{R}$. Compute

$$(\lambda \otimes \lambda)(A \times B).$$

Solution 70/107 – Product measure of a rectangle

Detailed solution

For measurable rectangles,

$$(\lambda \otimes \lambda)(A \times B) = \lambda(A)\lambda(B).$$

Now

$$\lambda([0, 2]) = 2, \quad \lambda((1, 4)) = 3.$$

Thus

$$(\lambda \otimes \lambda)(A \times B) = 2 \cdot 3 = 6.$$

Exercise 71/107 – Double integral on a rectangle

Problem

Difficulty: ★★

Compute

$$\int_0^1 \int_0^2 (x + y) dy dx.$$

Solution 71/107 – Double integral on a rectangle

Detailed solution

First integrate with respect to y :

$$\int_0^2 (x + y) dy = \int_0^2 x dy + \int_0^2 y dy.$$

This gives

$$2x + \left[\frac{y^2}{2} \right]_0^2 = 2x + 2.$$

Now integrate over x :

$$\int_0^1 (2x + 2) dx = \left[x^2 + 2x \right]_0^1 = 1 + 2 = 3.$$

Therefore the double integral equals 3.

Exercise 72/107 – Switching order on a triangle

Problem

Difficulty: ★ ★ ★

Compute

$$\int_0^1 \int_0^x y \, dy \, dx$$

and rewrite the integral with the order reversed.

Solution 72/107 – Switching order on a triangle

Detailed solution

Direct computation:

$$\int_0^x y \, dy = \frac{x^2}{2}.$$

Then

$$\int_0^1 \frac{x^2}{2} \, dx = \frac{1}{6}.$$

The domain is

$$0 \leq y \leq x \leq 1.$$

Reversing the order: $0 \leq y \leq 1$ and $y \leq x \leq 1$. Thus

$$\int_0^1 \int_0^x y \, dy \, dx = \int_0^1 \int_y^1 y \, dx \, dy.$$

The reversed integral gives

$$\int_0^1 y(1-y) \, dy = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Exercise 73/107 – Tonelli with a nonnegative function

Problem

Difficulty: ★ ★ ★

Let

$$f(x, y) = \frac{1}{(1 + x + y)^2}$$

on $(0, \infty)^2$. Can Tonelli's theorem be applied? What does it allow?

Solution 73/107 – Tonelli with a nonnegative function

Detailed solution

The function is nonnegative and measurable. Tonelli's theorem applies even before knowing whether the integral is finite. It allows us to write

$$\int_{(0,\infty)^2} f \, d(x,y) = \int_0^\infty \int_0^\infty \frac{1}{(1+x+y)^2} \, dy \, dx$$

with value possibly $+\infty$. Compute the inner integral:

$$\int_0^\infty \frac{1}{(1+x+y)^2} \, dy = \frac{1}{1+x}.$$

Then

$$\int_0^\infty \frac{1}{1+x} \, dx = \infty.$$

So the double integral is infinite.

Exercise 74/107 – Fubini with absolute integrability

Problem

Difficulty: ★ ★ ★

Show that Fubini applies to

$$f(x, y) = e^{-x} e^{-2y}$$

on $(0, \infty)^2$, and compute the double integral.

Solution 74/107 – Fubini with absolute integrability

Detailed solution

The function is nonnegative, so Tonelli applies. It is also integrable because

$$\int_0^{\infty} e^{-x} dx = 1, \quad \int_0^{\infty} e^{-2y} dy = \frac{1}{2}.$$

Since the function factors,

$$\int_0^{\infty} \int_0^{\infty} e^{-x} e^{-2y} dy dx = \left(\int_0^{\infty} e^{-x} dx \right) \left(\int_0^{\infty} e^{-2y} dy \right).$$

Thus the value is

$$1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Exercise 75/107 – Diagonal has zero area

Problem

Difficulty: ★ ★ ★

Let

$$D = \{(x, y) \in [0, 1]^2 : x = y\}.$$

Show that $(\lambda \otimes \lambda)(D) = 0$.

Solution 75/107 – Diagonal has zero area

Detailed solution

For each fixed $x \in [0, 1]$, the vertical section is

$$D_x = \{y : (x, y) \in D\} = \{x\}.$$

Thus

$$\lambda(D_x) = 0$$

for every x . By Tonelli applied to the indicator $\mathbf{1}_D$,

$$(\lambda \otimes \lambda)(D) = \int_0^1 \lambda(D_x) dx = \int_0^1 0 dx = 0.$$

The diagonal has infinitely many points but zero two-dimensional area.

Exercise 76/107 – Discrete Fubini

Problem

Difficulty: ★★

Let

$$a_{ij} = 2^{-i}3^{-j}, \quad i, j \geq 1.$$

Compute

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij}.$$

Solution 76/107 – Discrete Fubini

Detailed solution

All terms are nonnegative, so Tonelli's theorem for series allows us to factor the sum:

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} 2^{-i} 3^{-j} = \left(\sum_{i=1}^{\infty} 2^{-i} \right) \left(\sum_{j=1}^{\infty} 3^{-j} \right).$$

Now

$$\sum_{i=1}^{\infty} 2^{-i} = 1, \quad \sum_{j=1}^{\infty} 3^{-j} = \frac{1/3}{1 - 1/3} = \frac{1}{2}.$$

Therefore the double sum equals $1/2$.

Exercise 77/107 – When order matters for nonabsolute series

Problem

Difficulty: ★ ★ ★★

Explain why Fubini requires absolute integrability for signed functions. Use the idea of a conditionally convergent double series.

Solution 77/107 – When order matters for nonabsolute series

Detailed solution

For signed functions, one cannot freely exchange integrals unless the absolute integral is finite:

$$\int |f| < \infty.$$

In discrete form, this means

$$\sum_{i,j} |a_{ij}| < \infty.$$

If only the positive and negative parts cancel conditionally, different orders of summation can lead to different values or divergence. Thus Tonelli is safe for nonnegative terms, while Fubini for signed functions needs absolute integrability. Exam message: before switching order for a signed function, check $\int |f| < \infty$ or justify by a theorem.

Exercise 78/107 – Marginal densities by integration

Problem

Difficulty: ★ ★ ★

Let

$$f(x, y) = 6xy \mathbf{1}_{(0,1)^2}(x, y).$$

Compute the x -marginal

$$g(x) = \int_0^1 f(x, y) dy.$$

Solution 78/107 – Marginal densities by integration

Detailed solution

For $x \notin (0, 1)$, $g(x) = 0$. For $x \in (0, 1)$,

$$g(x) = \int_0^1 6xy \, dy = 6x \int_0^1 y \, dy.$$

Since

$$\int_0^1 y \, dy = \frac{1}{2},$$

we get

$$g(x) = 3x, \quad 0 < x < 1.$$

Thus

$$g(x) = 3x \mathbf{1}_{(0,1)}(x).$$

Exercise 79/107 – Product of simple functions

Problem

Difficulty: ★★

Let $f = 2\mathbf{1}_A$ and $g = 5\mathbf{1}_B$, where $\mu(A) = 3$ and $\nu(B) = 4$. Compute

$$\int_{X \times Y} f(x)g(y) d(\mu \otimes \nu)(x, y).$$

Solution 79/107 – Product of simple functions

Detailed solution

The product is

$$f(x)g(y) = 10\mathbf{1}_A(x)\mathbf{1}_B(y) = 10\mathbf{1}_{A \times B}(x, y).$$

Thus

$$\int_{X \times Y} f(x)g(y) d(\mu \otimes \nu) = 10(\mu \otimes \nu)(A \times B).$$

For a measurable rectangle,

$$(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B) = 3 \cdot 4 = 12.$$

Therefore the integral is

$$10 \cdot 12 = 120.$$

Module 2

L^p spaces and inequalities

Exercise 80/107 – Compute an L^p norm

Problem

Difficulty: ★

On $[0, 2]$, compute

$$\left\| 3\mathbf{1}_{[0,1]} \right\|_p$$

for $1 \leq p < \infty$.

Solution 80/107 – Compute an L^p norm

Detailed solution

By definition,

$$\|f\|_p = \left(\int |f|^p dx \right)^{1/p}.$$

Here $f = 3\mathbf{1}_{[0,1]}$, so

$$|f|^p = 3^p \mathbf{1}_{[0,1]}.$$

Therefore

$$\int_0^2 |f|^p dx = 3^p \lambda([0, 1]) = 3^p.$$

Taking the p -th root,

$$\boxed{\|3\mathbf{1}_{[0,1]}\|_p = 3.}$$

Exercise 81/107 – Essential supremum

Problem

Difficulty: ★★

On $[0, 1]$, define

$$f(x) = \begin{cases} 100, & x = 1/2, \\ 2, & x \neq 1/2. \end{cases}$$

Compute $\|f\|_\infty$.

Solution 81/107 – Essential supremum

Detailed solution

The ordinary supremum is 100, but the value 100 occurs only at one point. A single point has Lebesgue measure zero. The essential supremum ignores changes on null sets. Since $f = 2$ almost everywhere,

$$\|f\|_{\infty} = \operatorname{ess\,sup} |f| = 2.$$

Thus

$$\boxed{\|f\|_{\infty} = 2.}$$

Exercise 82/107 – L^p membership near zero

Problem

Difficulty: ★ ★ ★

For which $p \geq 1$ does

$$f(x) = x^{-1/3} \mathbf{1}_{(0,1)}(x)$$

belong to $L^p(0,1)$?

Solution 82/107 – L^p membership near zero

Detailed solution

We need

$$\int_0^1 |f(x)|^p dx < \infty.$$

Here

$$|f(x)|^p = x^{-p/3}.$$

The integral

$$\int_0^1 x^{-p/3} dx$$

converges if and only if

$$\frac{p}{3} < 1.$$

Therefore

$$f \in L^p(0, 1) \iff 1 \leq p < 3.$$

Exercise 83/107 – L^p membership at infinity

Problem

Difficulty: ★★

For which $p \geq 1$ is

$$f(x) = \frac{1}{1+x}$$

in $L^p(0, \infty)$?

Solution 83/107 – L^p membership at infinity

Detailed solution

We need

$$\int_0^{\infty} \left(\frac{1}{1+x} \right)^p dx < \infty.$$

Near 0, the function is bounded, so there is no problem. At infinity,

$$(1+x)^{-p} \sim x^{-p}.$$

The integral

$$\int_1^{\infty} x^{-p} dx$$

converges if and only if $p > 1$. Thus

$$f \in L^p(0, \infty) \iff p > 1.$$

Exercise 84/107 – Same L^p element

Problem

Difficulty: ★★

Let $f, g \in L^p([0, 1])$ and suppose $f = g$ except at finitely many points. Show that they define the same element of L^p .

Solution 84/107 – Same L^p element

Detailed solution

If f and g differ only at finitely many points, then

$$N = \{x : f(x) \neq g(x)\}$$

is finite. A finite set has Lebesgue measure zero. Thus

$$f = g \quad \text{almost everywhere.}$$

In L^p spaces, functions that are equal almost everywhere are identified. Equivalently,

$$\|f - g\|_p^p = \int |f - g|^p = 0.$$

Therefore they represent the same L^p element.

Exercise 85/107 – Inclusion on finite measure spaces

Problem

Difficulty: ★ ★ ★

Assume $\mu(X) < \infty$ and $1 \leq q < p < \infty$. Prove that

$$L^p(X) \subset L^q(X).$$

Solution 85/107 – Inclusion on finite measure spaces

Detailed solution

Let $f \in L^p$. We want to show $\int |f|^q < \infty$. Use the decomposition

$$X = \{|f| \leq 1\} \cup \{|f| > 1\}.$$

On $\{|f| \leq 1\}$, $|f|^q \leq 1$, so

$$\int_{\{|f| \leq 1\}} |f|^q d\mu \leq \mu(X) < \infty.$$

On $\{|f| > 1\}$, since $q < p$, $|f|^q \leq |f|^p$, so

$$\int_{\{|f| > 1\}} |f|^q d\mu \leq \int |f|^p d\mu < \infty.$$

Therefore $f \in L^q$.

Exercise 86/107 – No inclusion on infinite measure spaces

Problem

Difficulty: ★ ★ ★★

Give an example of a function in $L^2(1, \infty)$ but not in $L^1(1, \infty)$.

Solution 86/107 – No inclusion on infinite measure spaces

Detailed solution

Take

$$f(x) = \frac{1}{x}, \quad x > 1.$$

First,

$$\int_1^\infty |f(x)|^2 dx = \int_1^\infty \frac{1}{x^2} dx = 1 < \infty.$$

So $f \in L^2(1, \infty)$. But

$$\int_1^\infty |f(x)| dx = \int_1^\infty \frac{1}{x} dx = \infty.$$

Thus $f \notin L^1(1, \infty)$. This shows that finite-measure assumptions matter.

Exercise 87/107 – Holder inequality for two-point space

Problem

Difficulty: ★ ★ ★

On $X = \{1, 2\}$ with counting measure, let $f = (2, 1)$ and $g = (3, 4)$. Verify Holder's inequality for $p = 2$ and $q = 2$.

Solution 87/107 – Holder inequality for two-point space

Detailed solution

Holder with $p = q = 2$ is Cauchy-Schwarz:

$$\sum_{i=1}^2 |f_i g_i| \leq \left(\sum_{i=1}^2 |f_i|^2 \right)^{1/2} \left(\sum_{i=1}^2 |g_i|^2 \right)^{1/2}.$$

Left side:

$$|2 \cdot 3| + |1 \cdot 4| = 6 + 4 = 10.$$

Right side:

$$\sqrt{2^2 + 1^2} \sqrt{3^2 + 4^2} = \sqrt{5} \cdot 5 = 5\sqrt{5}.$$

Since $10 \leq 5\sqrt{5}$ is equivalent to $2 \leq \sqrt{5}$, the inequality holds.

Exercise 88/107 – Minkowski on a simple example

Problem

Difficulty: ★ ★ ★

On $X = \{1, 2, 3\}$ with counting measure, let

$$f = (1, 2, 0), \quad g = (0, 1, 2).$$

Verify Minkowski's inequality in L^2 .

Solution 88/107 – Minkowski on a simple example

Detailed solution

Compute

$$f + g = (1, 3, 2).$$

Then

$$\|f + g\|_2 = \sqrt{1^2 + 3^2 + 2^2} = \sqrt{14}.$$

Also

$$\|f\|_2 = \sqrt{1^2 + 2^2 + 0^2} = \sqrt{5}, \quad \|g\|_2 = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}.$$

Minkowski says

$$\sqrt{14} \leq 2\sqrt{5}.$$

Squaring gives $14 \leq 20$, which is true.

Exercise 89/107 – Cauchy-Schwarz for integrals

Problem

Difficulty: ★ ★ ★

Use Cauchy-Schwarz to prove

$$\left| \int_0^1 f(x) dx \right| \leq \|f\|_2$$

for every $f \in L^2(0, 1)$.

Solution 89/107 – Cauchy-Schwarz for integrals

Detailed solution

Write the integral as an inner product with the constant function 1:

$$\int_0^1 f(x) dx = \int_0^1 f(x) \cdot 1 dx.$$

By Cauchy-Schwarz,

$$\left| \int_0^1 f(x) \cdot 1 dx \right| \leq \left(\int_0^1 |f(x)|^2 dx \right)^{1/2} \left(\int_0^1 1^2 dx \right)^{1/2}.$$

The second factor is 1. Therefore

$$\left| \int_0^1 f(x) dx \right| \leq \|f\|_2.$$

Exercise 90/107 – Norm of an indicator

Problem

Difficulty: ★★

Let A be measurable with $0 < \mu(A) < \infty$. Compute

$$\|\mathbf{1}_A\|_p$$

for $1 \leq p < \infty$.

Solution 90/107 – Norm of an indicator

Detailed solution

By definition,

$$\|\mathbf{1}_A\|_p = \left(\int |\mathbf{1}_A|^p d\mu \right)^{1/p}.$$

Since $|\mathbf{1}_A|^p = \mathbf{1}_A$,

$$\int |\mathbf{1}_A|^p d\mu = \int \mathbf{1}_A d\mu = \mu(A).$$

Therefore

$$\boxed{\|\mathbf{1}_A\|_p = \mu(A)^{1/p}.$$

Exercise 91/107 – Essentially bounded but pointwise unbounded

Problem

Difficulty: ★ ★ ★

Define f on $[0, 1]$ by

$$f(0) = 10^6, \quad f(x) = 1 \text{ for } x \neq 0.$$

Is $f \in L^\infty([0, 1])$? Compute its L^∞ norm.

Solution 91/107 – Essentially bounded but pointwise unbounded

Detailed solution

Yes. The function differs from the constant function 1 only at the single point $\{0\}$. Since $\lambda(\{0\}) = 0$, we have

$$f = 1 \quad \text{almost everywhere.}$$

Therefore the essential supremum is

$$\operatorname{ess\,sup} |f| = 1.$$

So

$$f \in L^\infty([0, 1]) \quad \text{and} \quad \|f\|_\infty = 1.$$

Exercise 92/107 – An unbounded function in every finite L^p

Problem

Difficulty: ★ ★ ★

Show that

$$f(x) = |\log x|$$

belongs to $L^p(0, 1)$ for every $1 \leq p < \infty$, but $f \notin L^\infty(0, 1)$.

Solution 92/107 – An unbounded function in every finite L^p

Detailed solution

The function is unbounded near 0, so it is not essentially bounded:

$$\|f\|_{\infty} = \infty.$$

For finite p , compute

$$\int_0^1 |\log x|^p dx.$$

Use the substitution $x = e^{-t}$, so $dx = -e^{-t} dt$ and $x : 0 \rightarrow 1$ corresponds to $t : \infty \rightarrow 0$. Then

$$\int_0^1 |\log x|^p dx = \int_0^{\infty} t^p e^{-t} dt < \infty.$$

Thus $f \in L^p(0, 1)$ for all finite p , but not in L^{∞} .

Exercise 93/107 – Orthogonality in L^2

Problem

Difficulty: ★★

In $L^2(0, 2\pi)$, show that $\sin x$ and $\cos x$ are orthogonal.

Solution 93/107 – Orthogonality in L^2

Detailed solution

The inner product is

$$\langle \sin x, \cos x \rangle = \int_0^{2\pi} \sin x \cos x \, dx.$$

Use

$$\sin x \cos x = \frac{1}{2} \sin(2x).$$

Then

$$\int_0^{2\pi} \sin x \cos x \, dx = \frac{1}{2} \int_0^{2\pi} \sin(2x) \, dx = 0.$$

Therefore the two functions are orthogonal in $L^2(0, 2\pi)$.

Exercise 94/107 – L^2 distance between two functions

Problem

Difficulty: ★★

Compute the $L^2(0, 1)$ distance between $f(x) = x$ and $g(x) = 1$.

Solution 94/107 – L^2 distance between two functions

Detailed solution

The distance is

$$\|f - g\|_2 = \left(\int_0^1 |x - 1|^2 dx \right)^{1/2}.$$

Since $|x - 1|^2 = (1 - x)^2$,

$$\int_0^1 (1 - x)^2 dx = \int_0^1 (1 - 2x + x^2) dx.$$

Compute:

$$\left[x - x^2 + \frac{x^3}{3} \right]_0^1 = 1 - 1 + \frac{1}{3} = \frac{1}{3}.$$

Thus

$$\|f - g\|_2 = \frac{1}{\sqrt{3}}.$$

Exercise 95/107 – Convergence in L^p of shrinking indicators

Problem

Difficulty: ★ ★ ★

On $[0, 1]$, let

$$f_n = \mathbf{1}_{(0, 1/n)}.$$

Show that $f_n \rightarrow 0$ in L^p for every $1 \leq p < \infty$.

Solution 95/107 – Convergence in L^p of shrinking indicators

Detailed solution

Compute the L^p norm:

$$\|f_n - 0\|_p^p = \int_0^1 |\mathbf{1}_{(0,1/n)}|^p dx.$$

Since $|\mathbf{1}_A|^p = \mathbf{1}_A$,

$$\|f_n\|_p^p = \lambda((0, 1/n)) = \frac{1}{n}.$$

Therefore

$$\|f_n\|_p = \left(\frac{1}{n}\right)^{1/p} \rightarrow 0.$$

Hence $f_n \rightarrow 0$ in L^p .

Exercise 96/107 – Pointwise convergence but not L^1 convergence

Problem

Difficulty: ★ ★ ★

On $[0, 1]$, let

$$f_n = n \mathbf{1}_{(0, 1/n)}.$$

Show that $f_n \rightarrow 0$ pointwise but not in L^1 .

Solution 96/107 – Pointwise convergence but not L^1 convergence

Detailed solution

As before, for each fixed $x \in [0, 1]$, eventually $x \notin (0, 1/n)$, so $f_n(x) \rightarrow 0$. Now compute the L^1 distance from zero:

$$\|f_n\|_1 = \int_0^1 n \mathbf{1}_{(0, 1/n)} dx.$$

This equals

$$n \cdot \frac{1}{n} = 1.$$

Since $\|f_n\|_1 = 1$ for every n , it does not converge to 0. Thus pointwise convergence does not imply L^1 convergence.

Exercise 97/107 – Dual exponent

Problem

Difficulty: ★★

Find the conjugate exponent q for $p = 3$ and write Holder's inequality in this case.

Solution 97/107 – Dual exponent

Detailed solution

Conjugate exponents satisfy

$$\frac{1}{p} + \frac{1}{q} = 1.$$

For $p = 3$,

$$\frac{1}{3} + \frac{1}{q} = 1 \Rightarrow \frac{1}{q} = \frac{2}{3} \Rightarrow q = \frac{3}{2}.$$

Holder's inequality becomes

$$\int |fg| d\mu \leq \left(\int |f|^3 d\mu \right)^{1/3} \left(\int |g|^{3/2} d\mu \right)^{2/3}.$$

Exercise 98/107 – Finite measure norm comparison

Problem

Difficulty: ★ ★ ★ ★

Let $\mu(X) = 4$ and $f \in L^2(X)$. Prove

$$\|f\|_1 \leq 2 \|f\|_2.$$

Solution 98/107 – Finite measure norm comparison

Detailed solution

Apply Cauchy-Schwarz to $|f| \cdot 1$:

$$\int |f| d\mu \leq \left(\int |f|^2 d\mu \right)^{1/2} \left(\int 1^2 d\mu \right)^{1/2}.$$

The first factor is $\|f\|_2$. The second factor is

$$\left(\int 1 d\mu \right)^{1/2} = \mu(X)^{1/2} = 4^{1/2} = 2.$$

Therefore

$$\|f\|_1 \leq 2 \|f\|_2.$$

Exercise 99/107 – Approximation by simple functions

Problem

Difficulty: ★★

Let $f(x) = x$ on $[0, 1]$. Define

$$s_n(x) = \sum_{k=1}^n \frac{k-1}{n} \mathbf{1}_{[(k-1)/n, k/n)}(x).$$

Show that $s_n(x) \uparrow x$ for every $x \in [0, 1)$.

Solution 99/107 – Approximation by simple functions

Detailed solution

For a fixed $x \in [0, 1)$, choose k such that

$$\frac{k-1}{n} \leq x < \frac{k}{n}.$$

Then

$$s_n(x) = \frac{k-1}{n} \leq x.$$

Also the mesh size is $1/n$, so

$$0 \leq x - s_n(x) < \frac{1}{n}.$$

Therefore $s_n(x) \rightarrow x$. A monotone increasing construction can be obtained by taking dyadic lower approximations. The main idea is that measurable nonnegative functions are limits of simple functions from below.

Module 2

Mixed exam-style review

Exercise 100/101 – Simple functions are dense in L^1 in a concrete case

Problem

Difficulty: ★★☆☆

Let $f(x) = x^2$ on $[0, 1]$. Explain how to construct simple functions s_n such that $s_n \rightarrow f$ in L^1 .

Detailed solution

Partition $[0, 1]$ into intervals

$$I_{k,n} = \left[\frac{k-1}{n}, \frac{k}{n} \right), \quad k = 1, \dots, n.$$

Define a step function by the left endpoint value:

$$s_n(x) = \sum_{k=1}^n \left(\frac{k-1}{n} \right)^2 \mathbf{1}_{I_{k,n}}(x).$$

Since $f(x) = x^2$ is uniformly continuous on $[0, 1]$, the maximum oscillation of f on each interval tends to 0. Thus

$$\sup_{x \in [0,1]} |s_n(x) - f(x)| \rightarrow 0.$$

Then

$$\|s_n - f\|_1 \leq \lambda([0, 1]) \|s_n - f\|_\infty \rightarrow 0.$$

Exercise 101/107 – Null sets and L^p norms

Problem

Difficulty: ★★

Let $N \subset [0, 1]$ have Lebesgue measure zero and let $f = \mathbf{1}_N$. Compute $\|f\|_p$ for $1 \leq p < \infty$.

Solution 101/107 – Null sets and L^p norms

Detailed solution

Since $f = \mathbf{1}_N$,

$$|f|^p = \mathbf{1}_N.$$

Therefore

$$\|f\|_p^p = \int_0^1 \mathbf{1}_N dx = \lambda(N) = 0.$$

Hence

$$\|f\|_p = 0.$$

In the L^p space, $\mathbf{1}_N$ is the same element as the zero function.

Exercise 102/107 – Measurability plus integrability

Problem

Difficulty: ★ ★ ★

Let

$$f(x) = \begin{cases} 1/x^{1/2}, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Show that f is measurable and decide whether $f \in L^1(\mathbb{R})$ and $f \in L^2(\mathbb{R})$.

Solution 102/107 – Measurability plus integrability

Detailed solution

The function is measurable because it is continuous on $(0, 1)$ and extended by 0 outside a Borel set. For L^1 :

$$\int_0^1 x^{-1/2} dx = 2 < \infty.$$

So $f \in L^1$. For L^2 :

$$\int_0^1 |f(x)|^2 dx = \int_0^1 x^{-1} dx = \infty.$$

So $f \notin L^2$. Thus $f \in L^1(\mathbb{R})$ but $f \notin L^2(\mathbb{R})$.

Exercise 103/107 – Choose the correct theorem

Problem

Difficulty: ★ ★ ★

Let

$$f_n(x) = \left(1 - \frac{1}{n}\right) \mathbf{1}_{[0,1]}(x).$$

Which theorem can be used to show convergence of the integrals, MCT or DCT?

Solution 103/107 – Choose the correct theorem

Detailed solution

The sequence is increasing because $1 - 1/n$ increases to 1. Thus MCT can be used for the nonnegative functions f_n . Also,

$$0 \leq f_n \leq 1$$

and $1 \in L^1([0, 1])$, so DCT can also be used. Both theorems give

$$\lim_{n \rightarrow \infty} \int_0^1 \left(1 - \frac{1}{n}\right) dx = \int_0^1 1 dx = 1.$$

Exam lesson: sometimes more than one theorem applies; always check hypotheses explicitly.

Exercise 104/107 – Do not use DCT without domination

Problem

Difficulty: ★ ★ ★ ★

Let

$$f_n(x) = \sqrt{n} \mathbf{1}_{(0,1/n)}(x)$$

on $[0, 1]$. Does $f_n \rightarrow 0$ in L^1 ? Does $f_n \rightarrow 0$ in L^2 ?

Solution 104/107 – Do not use DCT without domination

Detailed solution

First, $f_n \rightarrow 0$ pointwise. For L^1 :

$$\|f_n\|_1 = \int_0^{1/n} \sqrt{n} \, dx = \sqrt{n} \frac{1}{n} = \frac{1}{\sqrt{n}} \rightarrow 0.$$

So $f_n \rightarrow 0$ in L^1 . For L^2 :

$$\|f_n\|_2^2 = \int_0^{1/n} n \, dx = n \frac{1}{n} = 1.$$

Thus $\|f_n\|_2 = 1$ for all n , so f_n does not converge to 0 in L^2 . Different norms see concentration differently.

Exercise 105/107 – A mixed sigma-algebra and integral problem

Problem

Difficulty: ★★ ★

Let $X = \{1, 2, 3, 4\}$, partitioned into atoms $P_1 = \{1, 2\}$ and $P_2 = \{3, 4\}$. Let $\mathcal{F} = \sigma(P_1, P_2)$ and counting measure μ . Which functions $f : X \rightarrow \mathbb{R}$ are $\mathcal{F}$ -measurable? Compute $\int f d\mu$ for $f(1) = f(2) = 3$, $f(3) = f(4) = 5$.

Solution 105/107 – A mixed sigma-algebra and integral problem

Detailed solution

A function is measurable with respect to the partition sigma-algebra exactly when it is constant on each atom. Indeed, points inside the same atom cannot be separated by measurable sets. Thus f is $\mathcal{F}$ -measurable if and only if

$$f(1) = f(2), \quad f(3) = f(4).$$

The given function satisfies this condition. With counting measure,

$$\int f \, d\mu = f(1) + f(2) + f(3) + f(4) = 3 + 3 + 5 + 5 = 16.$$

Equivalently, $f = 3\mathbf{1}_{P_1} + 5\mathbf{1}_{P_2}$ and $\mu(P_1) = \mu(P_2) = 2$.

Exercise 106/107 – Full proof: integral zero implies zero a.e.

Problem

Difficulty: ★★★★★

Let $f \geq 0$ be measurable. Prove that if

$$\int f \, d\mu = 0,$$

then $f = 0$ almost everywhere.

Solution 106/107 – Full proof: integral zero implies zero a.e.

Detailed solution

For each $n \geq 1$, define

$$A_n = \{x : f(x) > 1/n\}.$$

On A_n , we have $f \geq 1/n$. Therefore

$$0 = \int f \, d\mu \geq \int_{A_n} f \, d\mu \geq \frac{1}{n} \mu(A_n).$$

Hence $\mu(A_n) = 0$ for every n . Now

$$\{f > 0\} = \bigcup_{n=1}^{\infty} \{f > 1/n\} = \bigcup_{n=1}^{\infty} A_n.$$

A countable union of null sets is null, so

$$\mu(\{f > 0\}) = 0.$$

Thus $f = 0$ almost everywhere.

Exercise 107/107 – Final mixed exam exercise

Problem

Difficulty: ★★★★★

On $[0, 1]$, let

$$f_n(x) = n^{1/3} \mathbf{1}_{(0, 1/n)}(x) + x^n.$$

Find the pointwise limit and decide whether $f_n \rightarrow 0$ in L^1 and in L^2 .

Solution 107/107 – Final mixed exam exercise

Detailed solution

For each fixed $x > 0$, eventually $x \notin (0, 1/n)$, so the spike term is 0. Also $x^n \rightarrow 0$ for $x \in [0, 1)$, while $x^n = 1$ at $x = 1$. The pointwise limit is 0 almost everywhere. For L^1 , estimate separately:

$$\left\| n^{1/3} \mathbf{1}_{(0, 1/n)} \right\|_1 = n^{1/3} \frac{1}{n} = n^{-2/3} \rightarrow 0,$$

and

$$\|x^n\|_1 = \int_0^1 x^n dx = \frac{1}{n+1} \rightarrow 0.$$

Thus $f_n \rightarrow 0$ in L^1 . For L^2 , the spike term has

$$\left\| n^{1/3} \mathbf{1}_{(0, 1/n)} \right\|_2^2 = n^{2/3} \frac{1}{n} = n^{-1/3} \rightarrow 0.$$

Also

$$\|x^n\|_2^2 = \int_0^1 x^{2n} dx = \frac{1}{2n+1} \rightarrow 0.$$

By Minkowski, $\|f_n\|_2 \rightarrow 0$. Hence $f_n \rightarrow 0$ in L^2 as well.

Before the exam

For Module 2, make sure you can do the following without looking at notes.

- ① Check sigma-algebra axioms and build generated sigma-algebras from atoms.
- ② Compute measures using additivity, subadditivity and continuity from below/above.
- ③ Prove measurability using preimages and closure operations.
- ④ Compute Lebesgue integrals of indicators, simple functions and basic singular functions.
- ⑤ Choose correctly between MCT, Fatou, DCT, Tonelli and Fubini.
- ⑥ Work with L^p norms, essential supremum, Holder and Minkowski inequalities.
- ⑦ Produce counterexamples when a theorem hypothesis is missing.